

Date: _____

September 28, 2026 · 5:00 – 7:30 PM · Kai Santos

NOTES

MECH 191 — Metallurgy · Week 6 Notes · fill in blanks & key terms as you go

01 Nonferrous Metals

02 Iron-Carbon Phase Diagram

03 Putting It Together

SLIDE 2 *Class Agenda*

EXAMET-5-R-600-HD • 50-500x

Specimens to Check Out

- Specimen G — 6061-T6 aluminum
- Specimen K — 70/30 brass

How to check out the kit:

- 1 See instructor after class today
- 2 Sign out the specimen tray
- 3 Use the EXAMET-5 at your convenience
- 4 Return kit before leaving class
- 5 Submit completed module with your portfolio

Portfolio Handout — Module 5

- SLIDE 3** *Metallography Lab — Module 5 EXAMET-5-R-600-HD · 50–500×*

MECH 191 — Course & Unit Roadmap

Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
Materials Fundamentals	Steel & Alloys	Testing & Quality	Manufacturing	Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	Wk 4 Sep 14 Carbon & Alloy Steels	Wk 7 Oct 5 Mechanical Testing	Wk 10 Oct 26 Casting	Wk 14 Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	Wk 5 Sep 21 Stainless & Cast Irons	Wk 8 Oct 12 NDT Methods	Wk 11 Nov 2 Forming	Wk 15 Nov 30 Weld Metallurgy & Failure
	Wk 6 ← here Sep 28 Nonferrous & Phase Diagrams	Wk 9 Oct 19 Quality Standards	Wk 12 Nov 9 Machining	Wk 16 Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 • FINAL EXAM

SLIDE 4 MECH 191 — Course & Unit Roadmap

NOTES

Topics

- Aluminum, copper, and titanium — why each family exists alongside steel
- Aluminum alloy designation (1xxx-7xxx) and heat-treatable vs. work-hardened series
- Brass and bronze — how copper alloys balance conductivity, strength, and corrosion resistance
- Titanium's alpha/beta microstructure and why Ti-6Al-4V dominates aerospace
- The iron-carbon phase diagram: ferrite, austenite, cementite, pearlite, and eutectoid
- CCT diagram — what actually forms during real cooling vs. equilibrium predictions
- Annealing, normalizing, quenching, tempering, and case hardening

Reference Material

Moniz Ch. 6 — Phase Diagrams

Kalpakjian Ch. 7
Nonferrous Metals and Alloys

Iron-Carbon Phase Diagram (Interactive)

Solidification & Dendrite Growth

Covers Weeks 4-6

NOTES

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

- ## SLIDE 6 Core Questions

Aluminum

- $\sim 2.7 \text{ g/cm}^3$ — dominant in aerospace, automotive, and structural applications where weight matters
- Naturally corrosion-resistant — stable Al_2O_3 passive layer (similar to stainless)
- 1xxx–7xxx designation system; heat-treatable (2xxx, 6xxx, 7xxx) and work-hardened (1xxx, 3xxx, 5xxx) series
- Melts at $\sim 660^\circ\text{C}$ — much lower than steel; weldable but requires different process (GTAW/GMAW)

- Standard for electrical wiring, busbars, and heat exchangers — no common structural metal matches it
- Brass (Cu-Zn): good strength, machinability, corrosion resistance — fittings, valves, decorative
- Bronze (Cu-Sn or Cu-Al/Si): higher strength and wear resistance — bearings, bushings, marine hardware
- High thermal conductivity demands greater heat input when welding compared to steel

- ~60% the density of steel, with comparable or superior strength — used in aerospace, medical implants
- Corrosion resistance exceeds even the best stainless in most environments
- Alpha (HCP), beta (BCC), and alpha-beta alloys — Ti-6Al-4V (Grade 5) is the workhorse alloy
- Extremely reactive at elevated temps — must weld in inert shielding or vacuum to prevent embrittlement

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4-Digit Designation System

Technician Notes

~660°C — far lower than steel. High heat input in welding can cause burn-through or hot cracking without proper technique.

GTAW (TIG) or GMAW (MIG) with inert shielding. AC polarity for GTAW to break up the tenacious Al_2O_3 surface oxide.

T6 temper (solution treat + age) is the most common condition for 6061 and 7075 — provides near-peak strength.

Aluminum is anodic relative to steel and copper. Avoid direct contact in wet environments; use isolation gaskets.

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Best electrical & thermal conductivity of structural metals

- Pure copper (C10000–C18000): highest conductivity — standard for wiring, busbars, heat exchangers
- Brass (Cu-Zn): improved strength and machinability over pure Cu — fittings, valves, fasteners, decorative parts
- Bronze (Cu-Sn): better wear resistance than brass — bearings, bushings, gear blanks, marine hardware
- Cu-Al / Cu-Si / Cu-Be bronzes: specialist alloys for springs, corrosion-resistant bolts, non-sparking tools
- Welding: high thermal conductivity demands greater heat input than steel; most grades weld with GTAW or GMAW; oxygen-free grades preferred to prevent porosity

Best strength-to-weight ratio · exceptional corrosion resistance

- Density $\sim 4.5 \text{ g/cm}^3$ (60% of steel) with comparable or superior strength — indispensable in aerospace
- Alpha alloys (HCP): best creep resistance at high temperature; weldable; CP titanium used in chemical processing
- Beta alloys (BCC): highest strength, heat-treatable; used for fasteners and springs
- Alpha-beta: Ti-6Al-4V (Grade 5) — accounts for $\sim 50\%$ of all titanium used; dominant in aircraft frames, jet engine fan blades, medical implants
- Welding: extremely reactive above 600°C — must use inert gas trailing shield or weld in a purge box/vacuum; any oxygen or nitrogen pickup causes brittle embrittlement

SLIDE 9 Copper & Titanium — Key Properties and Alloy Families

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What It Tells You

- ## Key Regions & Points

BCC, max 0.022% C at 727°C — soft and ductile; stable phase in low-carbon steels at room temperature

FCC, dissolves up to 2.14% C at 1147°C — the phase steel must be in for most heat treatment to work

6.67% C — extremely hard, brittle iron carbide; forms when carbon exceeds solubility limit

Not a phase — a microstructural constituent: lamellar ferrite + cementite formed at eutectoid (0.77% C, 727°C)

0.77% C at 727°C — austenite transforms simultaneously to ferrite + cementite: boundary between hypo- and hypereutectoid

2.14% C = boundary — above this, carbon can no longer fully dissolve in austenite

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$< 0.77\% \text{ C}$

- On slow cooling from austenite, proeutectoid ferrite forms first at grain boundaries
- Remaining austenite then transforms to pearlite at 727°C
- Room-temperature microstructure: ferrite + pearlite
- More carbon → more pearlite → higher strength, lower ductility
- Examples: 1020 (0.20% C), 1040 (0.40% C) — most structural steels fall here

$$= 0.77\% \text{ C}$$

- Austenite transforms entirely to pearlite at exactly 727°C — no proeutectoid phase
- Finest microstructure achievable by slow cooling alone
- Maximum hardness from slow cooling, but martensite gives far higher hardness when quenched
- Example: 1080 (0.80% C) — wire rope, rail steel

> 0.77% C

- On slow cooling, proeutectoid cementite forms at prior austenite grain boundaries before pearlite
- Cementite network can cause brittleness — often spheroidize annealed to break up the network
- Room-temperature microstructure: cementite + pearlite
- Examples: 1095 (0.95% C), 52100 bearing steel (1.0% C) — tools, springs, bearings

SLIDE 11 *Reading the Phase Diagram — Steel Classification*

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The Technician's Perspective — Putting It Together

Material Selection	• Use aluminum (6061-T6) where weight matters and strength is moderate — most structural fabrication, frames, enclosures • Specify copper alloy based on function: pure Cu for conductivity; brass for fittings/machinability; bronze for bearings/wear • Titanium when steel is too heavy AND stainless isn't corrosion-resistant enough — budget for specialized welding procedures
Reading a WPS / Material Cert	• Phase diagram knowledge lets you interpret weld heat-affected zone behavior — why HAZ can soften (Al) or harden (medium-C steel) • CCT diagram reasoning: if you see 'preheat to 200°C,' that's slowing the cooling rate to avoid martensite in the HAZ • Temper color on stainless or Ti welds tells you the surface temperature reached — purple = ~300°C, straw = ~200°C
Heat Treatment Specifications	• 'Normalize and temper' on a drawing means two separate heating cycles — not one • Tempering temperature is critical: 150°C gives a different hardness than 450°C; verify with hardness test after • Case depth on carburized parts is specified on drawing — confirm by sectioning a coupon and etching with nital

SLIDE 12 *The Technician's Perspective — Putting It Together*

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Core Questions — Answers

What are the three major nonferrous families and what unique advantage does each offer?

Aluminum (one-third the density of steel — weight-critical apps), **Copper** (best electrical/thermal conductivity — wiring & heat exchangers), **Titanium** (best strength-to-weight ratio + exceptional corrosion resistance — aerospace & medical).

What does the iron-carbon phase diagram tell you — and what does it NOT tell you?

It maps which phases are stable at equilibrium for any temperature and carbon %. It does NOT show what forms during real industrial cooling — that requires the CCT diagram, which accounts for cooling rate.

What is the difference between the iron-carbon phase diagram and a CCT diagram?

Phase diagram = thermodynamic equilibrium — what forms if you hold temperature long enough. CCT = continuous cooling — what actually forms at different cooling rates. CCT is steel-specific; phase diagram covers the whole Fe-C system.

How does cooling rate determine the microstructure — and properties — of steel?

Slow → carbon diffuses fully → soft pearlite. Intermediate → bainite (stronger, tougher). Fast quench → carbon trapped → hard brittle martensite. CCT diagram shows exactly which cooling curve produces which result.

What does tempering do to as-quenched martensite, and what trade-off does it involve?

Tempering reheats martensite below the austenite region, letting some carbon diffuse and relieving internal stress. Trade-off: higher tempering temperature = more toughness but lower hardness. Required because as-quenched martensite is too brittle for service.

SLIDE 13 *Core Questions — Answers*

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01

02

03

SLIDE 14 *Key Takeaways — Week 6*

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